

LZ fit writeup

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1 Problem setup

We have an annealing Hamiltonian as a function of time

$$H(s) = sH_p(s) + (1-s)H_p + (1-s)H_d + s(1-s)H_c \quad (1)$$

where H_p is the problem Hamiltonian, H_d is the driver Hamiltonian, H_c is the catalyst and $s = t/T_A$ where T_A is the anneal time and t . Explicitly,

$$H_d = -\sum_{i=1}^n \sigma_i^x, \quad H_p = \sum_{i \in V} h_i \sigma_i^z + \sum_{(i,j) \in E} J_{zz} \sigma_i^z \sigma_j^z, \quad H_c = J_{xx} \sigma_1^x \sigma_1^x \quad (2)$$

where $\Sigma_{i \in V}$ is indexed over the vertices of the graph (spins) and $\Sigma_{i,j \in E}$ is indexed over the edges of the graph (couplings between spins). σ_i^a denotes the set of Pauli spin matrices that act on the on the spin associated with the i th Hilbert space, constructed via a tensor product.

For the n -spin WMIS problem, where n is an odd integer, we observed in simulations that the spectrum of $H(s)$ has 2 points where the ground state energy and first excited state energy, $E_0(s)$ and $E_1(s)$ respectively, approach near-equal values. We shall label these points s_0 and s'_0 , where the latter is associated with the perturbative crossing, and the former with the tunable gap (the point of interest).

2 Landau-Zener theory

To apply a Landau-Zener exponential fit to our results, we need to make 2 initial approximations:

1. We only consider the subspace of the ground state energy and first excited state energy, reducing to a 2-state system. In practice, the Hamiltonian reduces from an $n \times n$ to a 2×2 matrix. This is again motivated by the simulation; for anneal times approximately greater than 10ns, negligible amplitude leaks to the higher energy eigenstates.

2. We expand $H(s)$ to first order around s_0 , which in the case of LZ transitions, is given by

$$H(s) = \bar{E}\mathbf{1} + \begin{pmatrix} A(s-s_0) & C \\ C & B(s-s_0) \end{pmatrix} + \mathcal{O}(s^2) \quad (3)$$

where $\mathbf{1}$ is the identity (2×2) and $\bar{E}$ is the average energy at the transition point, $\frac{1}{2}(E_0(s_0) + E_1(s_0))$. Let $\bar{H}(s) = H(s) - \bar{E}$ and $X = s - s_0^1$, such that

$$\bar{H}(X) = \begin{pmatrix} AX & C \\ C & BX \end{pmatrix} + \mathcal{O}(X^2), \quad (4)$$

our aim is to find expressions for A , B and C in terms of the values of the spectrum that we know. N.B. since we constructed $H(s)$ using only x and z Pauli matrices, which are real-valued, A , B and C are consequently real.

We now proceed with only the first order matrix in X and explicitly compute its eigenvalues,

$$E_{0,1}^{LZ}(X) = \frac{1}{2} \left((A+B)X \pm \sqrt{(A-B)^2 X^2 + 4C^2} \right), \quad (5)$$

where the superscript indicate that these are the LZ fits for the ground state energy and first excited state energy, indeed distinct from $E_0(s)$ and $E_1(s)$. To compute A , B and C , we require 3 equations, which are obtained by computing $E_{0,1}^{LZ}$, as well as the first and second derivative evaluated them at $X = 0$, such that

$$E_i^{LZ}(0) = (-1)^{i+1}C, \quad (6)$$

$$\left. \frac{dE_i^{LZ}}{dX} \right|_{X=0} = \frac{(A+B)}{2}, \quad (7)$$

$$\left. \frac{d^2 E_i^{LZ}}{dX^2} \right|_{X=0} = (-1)^{i+1} \frac{(A-B)^2}{4C}, \quad (8)$$

where $i \in \{0, 1\}$. Proceeding with the 3 equations for the ground state², simplifying to primed derivative notation and dropping the $X = 0$ label,

$$A = E_i'^{LZ} + \sqrt{E_i^{LZ} E_i''^{LZ}}, \quad B = E_i'^{LZ} - \sqrt{E_i^{LZ} E_i''^{LZ}}, \quad C = (-1)^{i+1} E_i. \quad (9)$$

¹This is a translation to a new coordinate system with $(s_0, \bar{E})$ at the origin, simplifying the following steps but having no effect on the final result.

²We may use either set of equations. The values indeed come out to be identical after explicit computation of A, B and C within numerical error.

A and B are on equal footing in equations 7 and 8, so it doesn't matter which sign we choose for the second term in the above expressions. Next, we need explicit expressions for $E_i'^{LZ}$ and $E_i''^{LZ}$. These are computed by finite differences using the computational data obtained from stepping s . The explicit expressions are standard results, namely central difference schemes, such that

$$E_i'^{LZ} = \frac{E_i(X+h) - E_i(X-h)}{2h} + \mathcal{O}(h^2) \quad (10)$$

and

$$E_i''^{LZ} = \frac{E_i(X+h) - 2E_i(X) + E_i(X-h)}{h^2} + \mathcal{O}(h^2) \quad (11)$$

where h is the width of the interval between adjacent points. We then take the average of A, B and C computed for either $E_i'^{LZ}$ and use these averages in the Landau-Zener fitting formula.

2.1 Diabatic transition probability

Having obtained the first order fit, we can compute the Landau-Zener probability, P_{LZ} which is the probability of ending up in the ground state following a diabatic transition, given by

$$P_{LZ} = \exp(-2\pi\Gamma), \quad (12)$$

with

$$\Gamma = \frac{(E_1^{LZ} - E_0^{LZ})^2}{4\hbar \left| \frac{\partial}{\partial s} (\tilde{E}_1 - \tilde{E}_0) \right|} T_A. \quad (13)$$

where the quantity $\frac{\partial}{\partial s} (\tilde{E}_1 - \tilde{E}_0)$ is the speed of approach of an avoided crossing.

In this case,

$$\tilde{E}_{0,1}^{LZ}(X) = \lim_{X \rightarrow \infty} E_{0,1}^{LZ}(X) = AX, BX. \quad (14)$$

Again, A and B are on equal footing in equation 13. Recall $E_1^{LZ} - E_0^{LZ} = 2C$, so

$$\Gamma = \frac{C^2}{B - A} T_A. \quad (15)$$